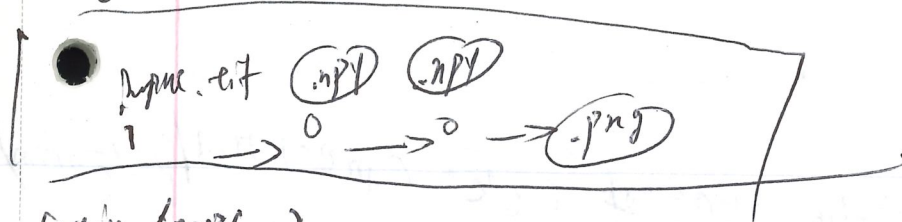


inginy



train-loop:

data load() → model() → forward() → pred/loss() → Backward

→ optimizer() → check-point() for long time training.  
adjust weights

when crashed, check-point() can  
recover/reload models/saves  
efficiently

precision goes low ⇒ training stage becomes  
numerically unstable.  
e.g. FP8

Mixed-precision-training()

⇒

Hardware sys design and model architecture design  
are synergistic

e.g.

Transformer trained on Nvidia GPU with ~~FP8~~ <sup>TF32</sup> FP8

would be faster than models trained on ~~other~~  
the same GPU with FP32.

# Lecture #3: Modern LLM Architecture.

original Transformer layer

variances:

architecture

norm in the front of block

(pre-norm/post-norm)

RoPE (position embeddings)

FF layers use SwiGLU, not ReLU

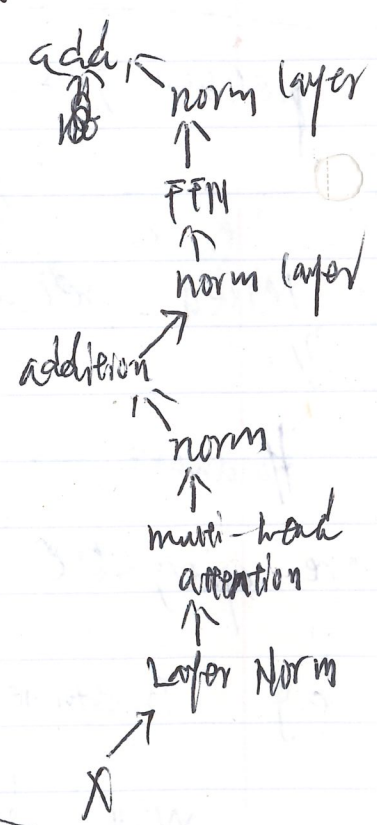
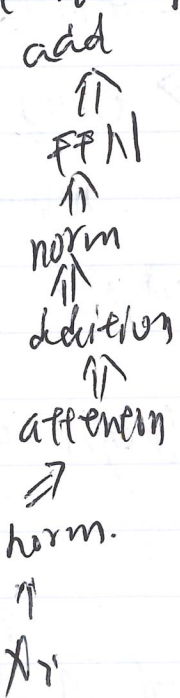
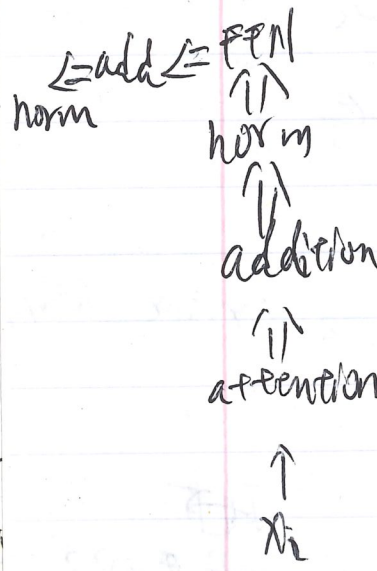
Attention variants

pre-norm

versus

post-norm

Versus: double norm



Layer Norm versus

RMS Norm

normalize mean and variance across dim

does not subtract mean or add a bias term

previous: GPT-3/2/1

LLM: faster/fewer operations